

VISTARA

INNOVERSE Demo Work Division

Purpose: Define the separate prototype, ML-demo, documentation and presentation work for the innovation competition. This workstream is independent of the main Vistara development project.

1. Demo Scope

The prototype focuses on one clear user journey rather than the complete Vistara platform.

Stay - Search, results and property details.

Smart Recommendation - Recommend suitable stays and explain why they match the user.

Property Verification - Show a verification workflow for the stay/property, including photo/property analysis as the ML concept.

Explore - Places, restaurants, experiences and local guidance in one destination experience.

Core demo journey

Home	Stay Search	Results	Smart Recommendation	Property Details	Verification	Explore
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2. Work Division

Area	Data Science / ML	Full Stack
Home & Search	Define search/recommendation inputs and data requirements.	Build home, navbar, destination search and filters.
Stay Results	Define ranking/recommendation logic and sample outputs.	Build result cards, sorting/filtering and result layout.
AI Recommendation	Build demo recommendation logic, output structure and reasons.	Integrate and display recommendations in the UI.
Property Details	Define ML-relevant property fields.	Build photos, amenities, reviews, location and booking-style details.
Property Verification	Define photo/property analysis concept and demo result.	Build verification status/workflow and verified badge.
Explore	Define data requirements for destination intelligence.	Build places, restaurants, experiences and local guidance.
ML API	Prepare FastAPI demo endpoints and contracts.	Connect UI/backend to the ML endpoints.
Demo Data	Prepare sample users, properties, reviews and ML examples.	Use the sample data throughout the prototype.
Presentation	Explain ML, intelligence and ML architecture.	Explain UI, system flow and user experience.

3. Data Science / ML Deliverables

Recommendation: Use a simple, explainable demo pipeline. Inputs may include location, budget, dates, guests, amenities and user preferences. Output should include property ID, rank/score and reasons.

User inputs → Feature preparation → Recommendation / ranking → property_id + score + rank + reasons

Property Verification: Focus on verifying the stay/property. The ML concept can analyze submitted property photos and related signals, returning a verification result for the application workflow.

Property information + photos → Image / property analysis → Verification signals → Verified / Needs Review

Minimum ML API:

POST /ml/recommend	Recommendation demo endpoint
POST /ml/image-verification	Property/photo verification demo endpoint
GET /ml/health	Service health check

4. Full Stack Deliverables

The Full Stack implementation should make the demo flow clickable and visually consistent. Priority screens are Home, Stay Search, Stay Results, Property Details and Explore.

The UI should visibly surface the two intelligent elements: **AI Recommended Stay** and **Verified Property**.

5. Explore Structure

Explore

Places	Restaurants	Experiences	Local Guidance
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Local guidance remains inside Explore. No separate Guidance or Guide page is required for the prototype.

6. Shared Work

Product design: Agree on the visual system, colors, typography, cards, navbar, images and overall premium presentation.

Integration: Keep the request/response contract between ML and the application fixed so the prototype is stable.

Presentation: Practice one continuous story instead of showing disconnected screens.

7. Recommended Demo Story

A user is visiting a destination.

01	Search	User searches for a stay.
02	Results	User reviews available properties.
03	Smart recommendation	Vistara shows the recommended stay and why it matches.
04	Property verification	User opens details and sees the property verification status.
05	Explore	User discovers places, food, experiences and local guidance.

8. Sprint Rules

1. Do not mix this repository with the main Vistara project.
2. Do not add large new features during the competition sprint.
3. Keep the ML scope small, visible and explainable.
4. Prioritize a smooth clickable demo over unfinished backend complexity.
5. Do not use real people's private data in demo datasets.
6. Keep secrets out of GitHub; use environment variables and .env.example.

9. Repository Separation

MAIN VISTARA	VISTARA-INNOVERSE
Main long-term project 90-day roadmap Phase 1 / Phase 2 / future development	Research Planning Prototype ML demo Demo data Presentation Documentation

10. Definition of Done

The prototype is ready when a judge can understand the concept from the UI without a long explanation, the complete demo flow works from start to finish, the recommendation feature is visible, property verification is understandable, Explore feels integrated, and both team members can explain their own contribution.

11. Final Scope

STAY + SMART RECOMMENDATION + PROPERTY VERIFICATION + EXPLORE